Generation
Screening
Modeling
Optimization
Validation
MatterGen structure generation
DFT-computed SSE labels
2
1
0
-1
-2
E − E_F (eV)
SSE
Γ
SSE: Max spin splitting in [E_F − 2 eV, E_F]
Binary altermagnet screening
Criterion
|M| < 0.005 μ_B
&
SSE ≥ 1 meV
322 parents
Strain sampling
± 5% strain
N = 3,845
Motif-based descriptors
Motif
MSBI
MPF
XMX_std
Z_M
52 DFT-free descriptors
↓
XGBoost surrogate for SSE
↓
SHAP interpretation
MSBI
MPF
p/d ratio
|SHAP value|
Descriptor-space Bayesian optimization
Optimization loop
Sampled descriptors
↓
XGBoost surrogate
↓
Maximize predicted SSE
↓
Top candidates
Prototype matching
CN = 4
CN = 6
DFT validation
Top-5 predicted candidates (CN ∈ {4, 6})
High SSE material α-NiS identified
0.0
−0.5
−1.0
−1.5
−2.0
E − E_F (eV)
L
Γ
L'
SSE: 0.823 eV